

# CHAPTER 7 : DIODES IN TELECOMMUNICATIONS

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\* Used to recover the Amplitude Modulated (AM) signal

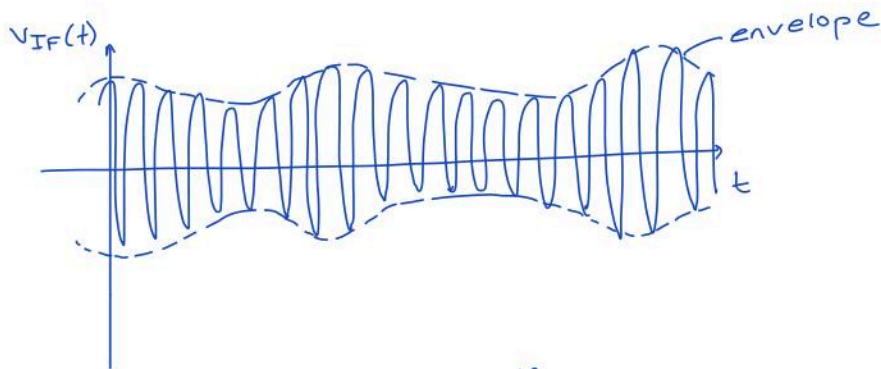
AM signal : at the output of the IF amplifier stage

$$V_{IF}(t) = V_c [1 + m(t)] \cos(\omega_{IF} t)$$

carrier amplitude

modulating signal (speech/music signal with ~3-4 kHz bandwidth)

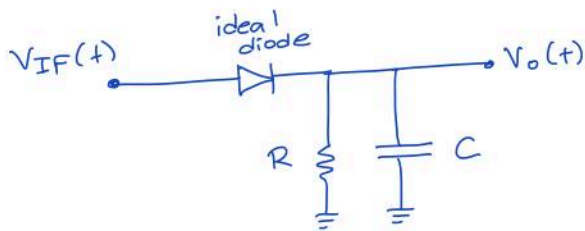
carrier freq. (@ 15 MHz)



Aim : AM demodulation

- determine  $m(t)$  from  $V_{IF}(t)$
- simplest method : "envelope detector"

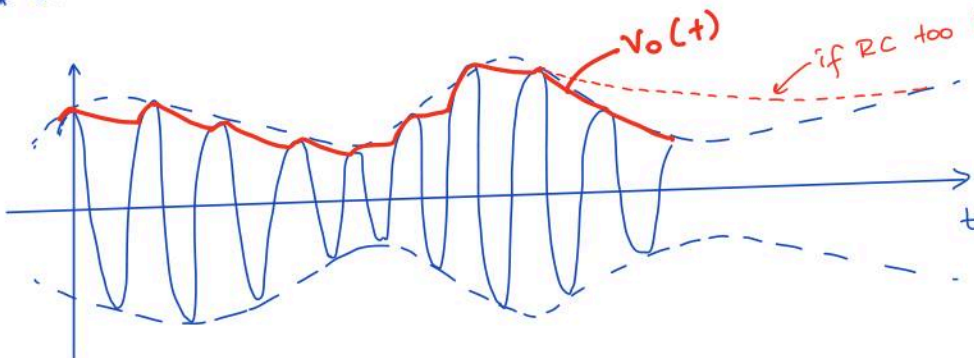
Envelope Detector : (same as half-wave rectifier circuit) but amplitude varies as a function of time



Diode only lets positive cycles pass, RC filters the signal.

\* when diode ON :  $V_O(t) = V_{IF}(t)$

\* when diode OFF : exponential decay with  $\tau = RC$



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Time constant RC : The capacitor should discharge fast enough to follow the envelope  
 $\Rightarrow$  need to look at the maximum negative slope of the envelope.

Let envelope :  $V_c \left[ 1 + \underbrace{\frac{V_m}{V_c} \cos(\omega_m t)}_{m(t)} \right]$ , where  $0 < \frac{V_m}{V_c} < 1$

slope :  $-V_m \omega_m \sin(\omega_m t)$

max. negative slope :  $-V_m 2\pi f_m$

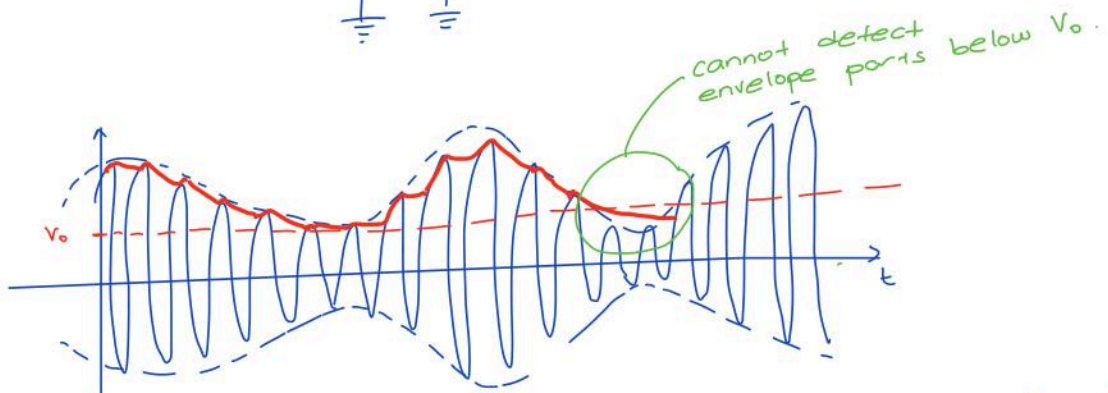
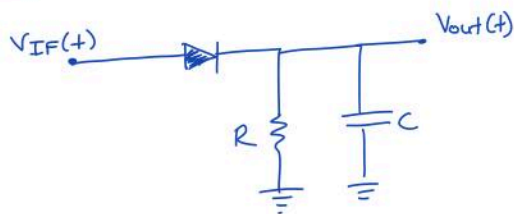
\* Compare with a decaying exponential :  $V_c \cdot e^{-t/RC}$   
 slope at  $t=0$  :  $-\frac{V_c}{RC}$

\* we want :  $\frac{V_c}{RC} > V_m 2\pi f_m \Rightarrow RC < \frac{V_c}{2\pi V_m f_m}$

\* if RC is too large : "failure-to-follow" distortion

\* Downside : Because RC cannot be large, we have ripples at the output.

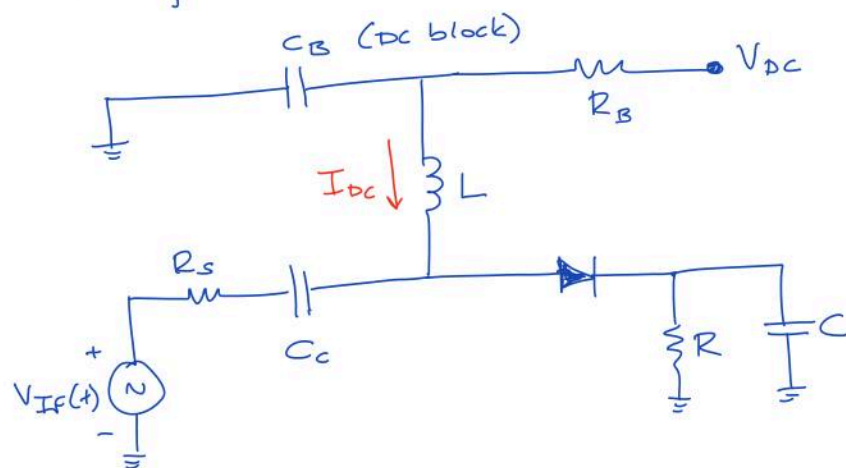
### Real Diodes in Envelope Detectors:



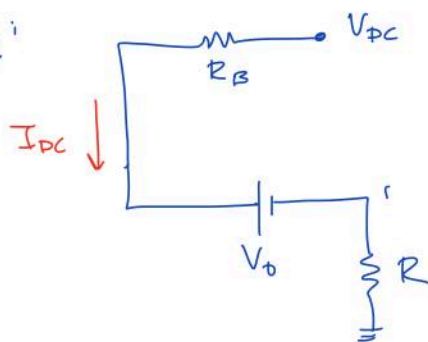
Finite turn on voltage causes distortions in the detected envelope.

Solution : Biasing :

\* We precondition the diode by forcing a DC current to flow through it.



DC response :



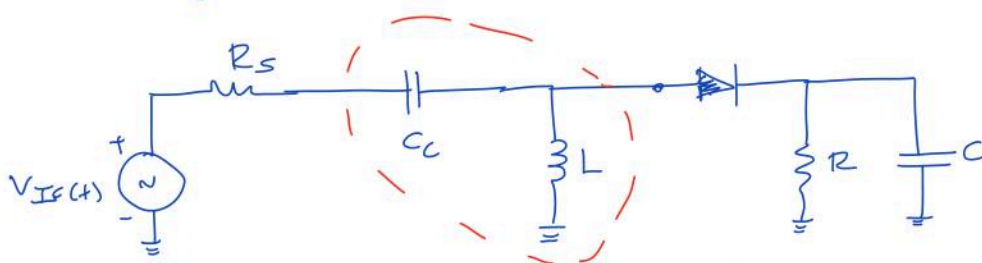
$$I_{DC} = \frac{V_{DC} - V_0}{R_B + R}$$

works like a current source  
large resistor

- \* use  $V_{DC} \gg V_0$ , and large  $R_B$  to make  $I_{DC}$  insensitive to  $V_0$  and internal diode resistance.
- \* Diode can be ON even when  $V_{IF}(t)$  is zero.
- \* The output voltage can follow the envelope down to very low positive input voltage levels.

AC response at IF freq :

$C_C$  &  $L$  are resonant at IF freq

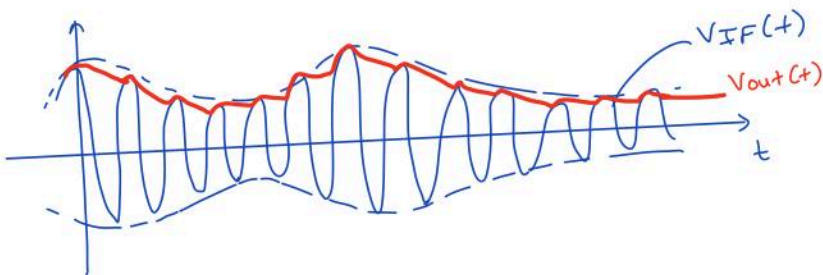


resonant circuit  
has BPF effect.  
Also boosts signal by Q factor  
of the resonant circuit

## Automatic Gain Control :

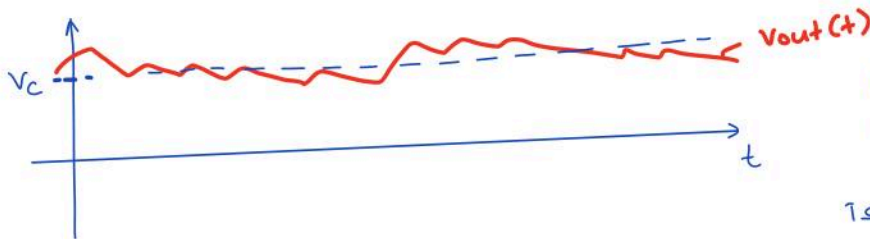
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- \* Depending on the distance between transmitter and receiver antennas,  $V_{IF}(t)$  will be scaled.
  - $\Rightarrow$  we need to set gain high if the signal is small
  - $\Rightarrow$  However, if signal is high, high gain may saturate the amplifiers.
- \* Automatic gain control is desirable.



$$V_{IF}(t) = \underline{V_c} [1 + m(t)] \cos(\omega_{IF} t)$$

$$V_{out}(t) \approx V_c (1 + m(t))$$



DC level of  $V_{out}(t)$  is equal to  $V_c$  (carrier amplitude), which is proportional to the signal level.

- \* Apply a LPF with very small cut-off frequency to extract DC voltage level only.

Large  $V_c \rightarrow$  small gain

Small  $V_c \rightarrow$  large gain

- \* We need a circuit element whose resistance changes in relation to a voltage.

## PIN Diode

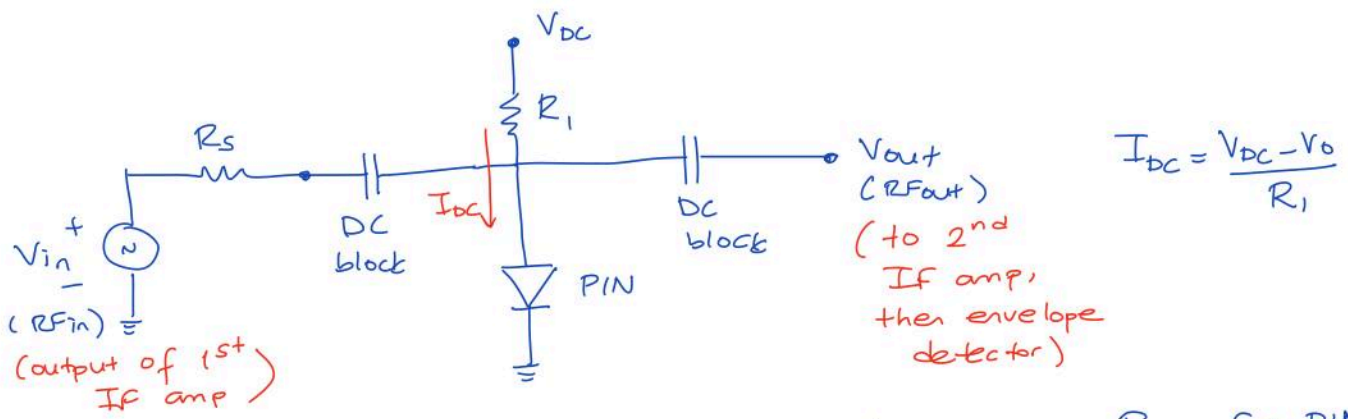
- p-type, intrinsic, n-type

- At frequencies  $> 5 \text{ MHz}$ , acts like a linear resistor

$$R_{RF} \approx \frac{K}{I_{DC}}$$

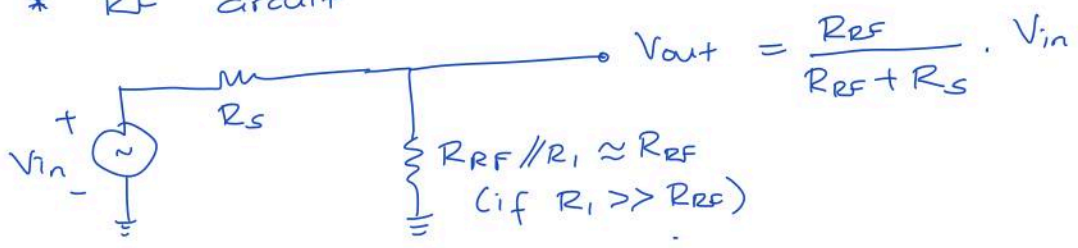
$\uparrow$  RF resistance depends on DC current  
 $\leftarrow$  DC current passing through the diode

- At low frequencies, acts like a regular diode

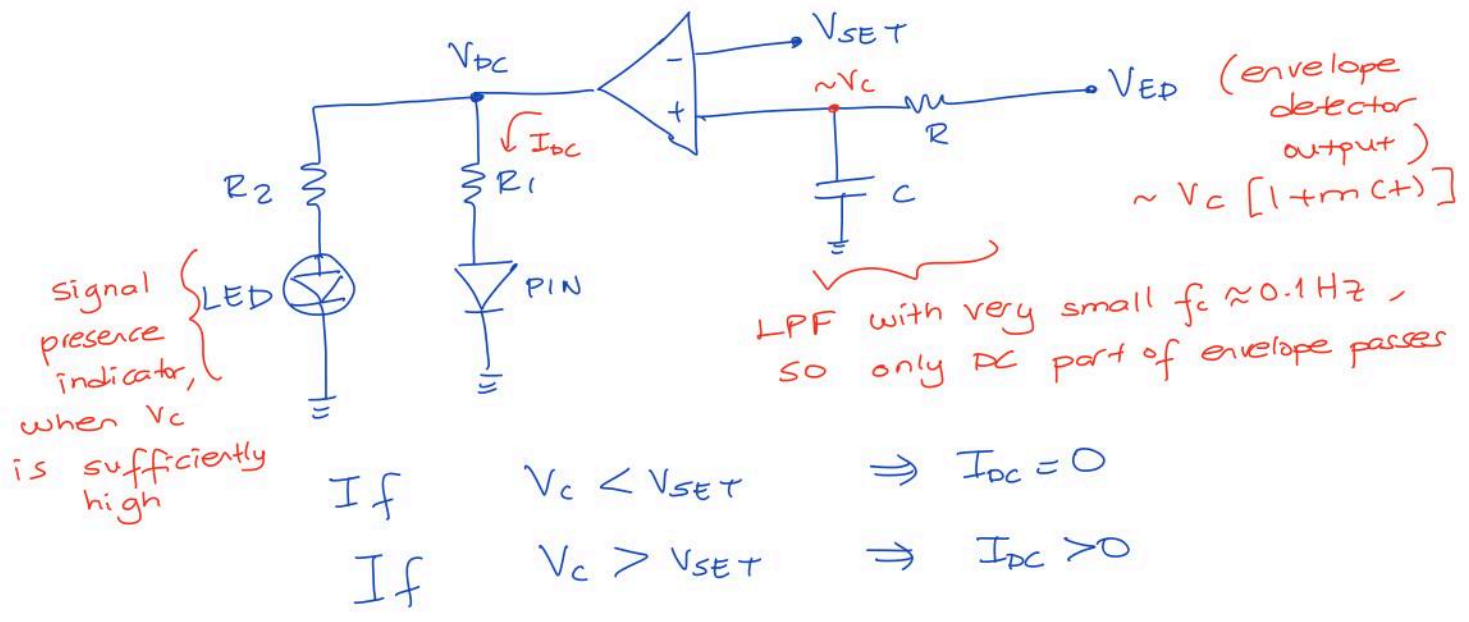


Change  $V_{DC}$  to change  $I_{DC} \Rightarrow$  changes  $R_{RF}$  for PIN

\* RF circuit



\* AGC input  $V_{DC}$ :



if  $V_c \uparrow$ ,  $I_{DC} \uparrow$ ,  $R_{RF} \downarrow \Rightarrow V_{out} \downarrow$ ,  $V_c \downarrow$

\* AGC only decreases the gain by reducing  $R_{RF}$  cannot increase gain

\* Overall circuit in TRC-II:

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